

SEEMAB — CLASS 9 (FEDERAL BOARD)

Mathematics exam-16 — Detailed Marking Report

Units 1, 2, 4, 5 and 7 • Revision Test 2026 • Attempted 21 August 2026

37/43

TOTAL

86.0%

PERCENTAGE

A

GRADE

20/24

SECTION A

8/9

SECTION B

9/10

SECTION C

SECTION A — Multiple Choice (20 / 24)

Twenty correct. The four below are the whole of the loss. Every one of the twenty-four questions was written against a stated Student Learning Outcome, and all four misses were re-checked against the printed option order after marking; all four are sound questions and nothing needed voiding.

Q5. The Babylonian system of numerals was a sexagesimal system, that is, a system with base:

She chose **A (10)**. Correct answer: **C (60)**. **-1**

“Sexagesimal” means base sixty. Her book says the Babylonian system “was a sexagesimal system of numbers (system with base 60)” and adds that the idea survives today in how we measure time and angles: 60 seconds in a minute, 60 minutes in an hour.

Source in her textbook: Unit 1, opening section, Babylonians’ Contribution to the Numeral System.

Q11. The factors of $27x^3 + 64$ are:

She chose **C $(3x - 4)(9x^2 + 12x + 16)$** . Correct answer: **A $(3x + 4)(9x^2 - 12x + 16)$** . **-1**

This is the same identity she missed on exam-15, and it is the one to fix before the next paper.

$27x^3 + 64 = (3x)^3 + 4^3$, a **sum** of cubes. The rule is $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$. She chose the option with the signs the other way round, which is the **difference** rule, $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$. The way to hold on to it: **the bracket sign matches the question, and the middle sign inside the trinomial is the opposite of it.**

Source in her textbook: Unit 4, §4.3, factorization of $a^3 + b^3$.

Q18. If $a < b$, then $a + c < b + c$ for every real number c . This is the:

She chose **C (transitive property)**. Correct answer: **A (additive property of inequality)**. **-1**

Something is being **added** to both sides, so it is the **additive** property. The transitive property is the chain rule: if $a < b$ and $b < c$ then $a < c$. Trichotomy is the one that says exactly one of $a < b$, $a = b$, $a > b$ is true.

Source in her textbook: Unit 5, §5.4, properties of inequalities.

Q22. If the ordered pairs $((a + 1)/2, b - 3)$ and $(4, 5)$ are equal, then the value of a is:

She chose **A (4)**. Correct answer: **B (7)**. **-1**

Equal ordered pairs means corresponding elements are equal, so $(a + 1)/2 = 4$. That gives $a + 1 = 8$ and **a** = 7. Option A is the value of the whole first element, not of a . Read what the question asks for: it wants **a**, not the coordinate.

Source in her textbook: Unit 7, Miscellaneous Exercise 7 Q1, equality of ordered pairs.

SECTION B — Short Questions (8 / 9)

Q.2(i) Laws of exponents, three parts **3 / 3**

(a) $x^5 \times x^{-2} = x^3$, then $x^3 \div x^4 = x^{-1} = 1/x$. (b) $(81)^{3/4} = (3^4)^{3/4} = 3^3 = 27$. (c) $\sqrt[4]{a^7} = a^{7/4}$. All three clean, with the working shown.

Q.2(ii) Evaluate $\log 48$, showing every step and naming the law used at each step 2 / 3

Her working, exactly as she wrote it:

$$\begin{aligned} &\log 48 \\ &\log 6 \times 8 \\ &\log 3 \times 2 \times 2 \times 2 \times 2 \\ &\log 3 + \log 2 + \log 2 + \log 2 + \log 2 \\ &0.4771 + 0.3010 + 0.3010 + 0.3010 + 0.3010 \\ &= 1.6811 \end{aligned}$$

The value is **exactly right** and the decomposition $48 = 3 \times 2 \times 2 \times 2 \times 2$ is a perfectly good route. The mark went on the part of the instruction she skipped: **she never named a single law**. Splitting $\log(3 \times 2 \times 2 \times 2 \times 2)$ into a sum of logs *is* the **product law**, and writing it as $4 \log 2$ would have been the **power law**. Naming them costs one line each.

This is now the third paper in a row where a mark hangs on the half of the instruction she did not answer (Mathematics exam-15 Q.2(iii), Chemistry exam-19 Q.3(i)(c), and now this). The maths is not the problem. Underline the verbs in the question before starting, and tick them off at the end.

Q.2(iii) Solve $|2z - 3| = 9$ and verify both solutions 3 / 3

Both cases taken, $z = 6$ and $z = -3$, and **both verified** by substitution. Full marks. Worth noting because the verification is exactly the extra step she left out on the log question: here she did it.

Small habit to correct, no mark lost: in the check she wrote " $2z - 3 = +9$ " rather than keeping the modulus bars, " $|2(6) - 3| = 9$ ". Keep the bars until the final line.

SECTION C — Long Questions (9 / 10)**Q.3(i)(a) Factorize completely: $1 + 4x^4$** 2 / 2

She spotted the $a^4 + 4b^4$ type, added and subtracted $4x^2$, reached $(1 + 2x^2)^2 - (2x)^2$ and finished at **$(2x^2 - 2x + 1)(2x^2 + 2x + 1)$** . Correct.

One presentational warning. Her second line reads " $(1)^2 + (2x^2)^2 + 2(1)(2x) - 2(1)(2x^2) - (2x^2)^2$ ", which has a term too many and is not a true identity as written. The lines after it are right, so no mark was lost, but a marker could stall on it. Write the add-and-subtract step as one clean line.

Q.3(i)(b) Find the square root of $49p^2 - 84pq + 36q^2$ by factorization 0 / 1

Her working, exactly as she wrote it:

$$\begin{aligned} (b) \quad &49p^2 - 84pq + 36q^2 \\ &(a-b)^2 = (a)^2 - 2(a)(b) + (b)^2 \\ &\quad \quad \quad a^2 - 2ab + b^2 \\ &(7p)^2 - 2(7p)(6q) + (6q)^2 \\ &(7p - 6q)^2 \end{aligned}$$

That last line is where she stopped. **Every step of the method is correct** — she recognised the perfect square and factorized it properly. But the question asked for the **square root**, and a square root never appeared on the page: no radical sign, no $\pm$, no final answer. She produced the square, not the root.

One more line was needed: **$\sqrt{49p^2 - 84pq + 36q^2} = \pm(7p - 6q)$** .

This is the same habit as Q.2(ii) above, in a different subject area. Two of the four marks she lost outside Section A on this paper are the last line of a question she had already done the hard part of.

Q.3(i)(c) Simplify the division of two algebraic fractions 2 / 2

Factorized all four expressions, including $x^2 + 11x + 24$ by splitting the middle term into $x^2 + 3x + 8x + 24$, inverted the second fraction, cancelled correctly and finished at **$x / (x - 8)$** , marked "Ans". This was the longest single piece of algebra on the paper and it is completely right.

Q.3(ii) Coordinate geometry, A(-1,-2), B(-1,4), C(7,4) 5 / 5

(a) $AB = 6$, $BC = 8$, $AC = \sqrt{100} = 10$, all by the distance formula. 2/2

(b) $AB^2 + BC^2 = 36 + 64 = 100 = AC^2$, then in her words “therefore, $\triangle ABC$ is a right triangle, and the longest side AC is the hypotenuse.” 1/1

(c) $M = (3, 1)$, then $MA = MB = MC = \sqrt{25} = 5$ units, all three computed separately. 2/2

A perfect five-mark answer, and the best work on the paper. She carried a multi-step proof through three different formulas without a slip.

Notation only: on the MB line she wrote $\sqrt{[(4)^2 + (9)]}$ instead of $\sqrt{[(4)^2 + (-3)^2]}$. The value is right, the line is not.

TWO THINGS FLAGGED, NEITHER COSTING A MARK

Q16 has a struck-out letter before the answer.

She wrote a letter, scored it out with several strokes, and wrote **D** beside it. On a written list her final intent is obvious and it was credited in full. But instruction 3 on the paper says “*Cutting or overwriting a bubble will not be credited*”, and in a real board exam with a bubble grid this would have scored **zero**. She needs to decide, then fill one bubble.

Q20’s letter is ambiguous in the scan.

The glyph could be read as a P. It was credited as **D**, which is correct, on two grounds: it is constructed identically to her looped D in Q4 and Q12, and P is not one of the four options. **If Seemab says she meant something else, this paper drops to 36/43 (83.7%).** Worth a ten-second check with her.

WHAT TO REVISE BEFORE THE NEXT MATHS PAPER

Item	Why it is on this list
$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$	Missed on two consecutive papers. exam-15 Q13 and now exam-16 Q11. Note that she then used the identity correctly inside Section C on exam-15, so this is recall of the sign pattern, not understanding.
Answer every verb in the question	Cost her the whole of her non-MCQ loss on this paper: naming the log laws (Q.2(ii)) and writing the square root (Q.3(i)(b)). Third paper running.
Properties of inequalities by name	Q18. Additive, multiplicative, transitive, trichotomy. She can apply them; she cannot yet name them on demand.
Equality of ordered pairs	Q22. Solve for the letter the question names, not for the whole element.
History of numbers, Unit 1	Q5. Sumerian, Babylonian sexagesimal (base 60), Egyptian. One page of her book, rarely revised.

— END OF MARKING REPORT —